```
clc;
clear;
close all;
```

INPUT PARAMETERS

```
fs_ecg = 360;                  % ECG sampling rate (Hz)
fs_eeg = 256;                  % EEG sampling rate (Hz)
fs_emg = 1000;                 % EMG sampling rate (Hz)

band_ecg = [0.5 40];           % ECG band (Hz)
band_eeg = [1 40];             % EEG band (Hz)
band_emg = [20 450];           % EMG band (Hz)

T = 10;                        % Acquisition duration (s)
mains = 50;                    % Power-line frequency (Hz)
filtOrder = 4;                 % Butterworth filter order

useRealData = false;           % true = load real datasets
ecgFile = 'ecg_data.mat';      % Contains ecg_raw
eegFile = 'eeg_data.mat';      % Contains eeg_raw
emgFile = 'emg_data.mat';      % Contains emg_raw

rng(1);                        % Repeatable synthetic noise
```

1. ACQUIRE / LOAD SIGNALS

```
if useRealData
    S = load(ecgFile);
    ecg_raw = S.ecg_raw(:)';

    S = load(eegFile);
    eeg_raw = S.eeg_raw(:)';

    S = load(emgFile);
    emg_raw = S.emg_raw(:)';

    % Keep only the first T seconds
    ecg_raw = ecg_raw(1:T*fs_ecg);
    eeg_raw = eeg_raw(1:T*fs_eeg);
    emg_raw = emg_raw(1:T*fs_emg);
else
    [ecg_raw, ecg_clean] = synth_ecg(fs_ecg, T, mains);
    [eeg_raw, eeg_clean] = synth_eeg(fs_eeg, T, mains);
    [emg_raw, emg_clean] = synth_emg(fs_emg, T, mains);
end

t_ecg = (0:numel(ecg_raw)-1)/fs_ecg;
t_eeg = (0:numel(eeg_raw)-1)/fs_eeg;
t_emg = (0:numel(emg_raw)-1)/fs_emg;
```

2. DIGITAL FILTERING

```
ecg_f = bp_filter(ecg_raw, band_ecg, fs_ecg, filtOrder);
eeg_f = bp_filter(eeg_raw, band_eeg, fs_eeg, filtOrder);
emg_f = bp_filter(emg_raw, band_emg, fs_emg, filtOrder);

% Add a notch filter to reduce 50 Hz hum in EMG
notchBand = [49 51] / (fs_emg/2);

[bn, an] = butter(2, notchBand, 'stop');

emg_f = filtfilt(bn, an, emg_f);
```

3. SPECTRA (FFT)

```
[fE1, XE1] = fft_spectrum(ecg_raw, fs_ecg);
[~, XE2] = fft_spectrum(ecg_f, fs_ecg);

[fG1, XG1] = fft_spectrum(eeg_raw, fs_eeg);
[~, XG2] = fft_spectrum(eeg_f, fs_eeg);

[fM1, XM1] = fft_spectrum(emg_raw, fs_emg);
[~, XM2] = fft_spectrum(emg_f, fs_emg);
```

4. FEATURE EXTRACTION

```
% ECG: R-peaks, RR interval, and heart rate
[pk, loc] = findpeaks(ecg_f, ...
    'MinPeakHeight', 0.5*max(ecg_f), ...
    'MinPeakDistance', round(0.3*fs_ecg));

rPeakTimes = loc/fs_ecg;
RR = diff(rPeakTimes);
HR_all = 60./RR;
HR_avg = 60/mean(RR);

% EEG: Band powers using Welch PSD
[Pxx, fw] = pwelch(eeg_f, hamming(2*fs_eeg), ...
    fs_eeg, 2*fs_eeg, fs_eeg);

bands = struct('name', {'Delta','Theta','Alpha','Beta'}, ...
    'range', {[1 4],[4 8],[8 13],[13 30]});

bandPower = zeros(1,4);

for k = 1:4
    idx = fw >= bands(k).range(1) & ...
          fw < bands(k).range(2);

    bandPower(k) = trapz(fw(idx), Pxx(idx));
end
```

```
relPower = 100*bandPower/sum(bandPower);
[~, domIdx] = max(bandPower);

% EMG: RMS amplitude and 100 ms sliding RMS envelope
RMS_raw = sqrt(mean(emg_raw.^2));
RMS_f = sqrt(mean(emg_f.^2));

win = round(0.1*fs_emg);
emg_env = sqrt(movmean(emg_f.^2, win));

% SNR estimate
SNR_ecg = snr_db(ecg_f, ecg_raw-ecg_f);
SNR_eeg = snr_db(eeg_f, eeg_raw-eeg_f);
SNR_emg = snr_db(emg_f, emg_raw-emg_f);
```

5. ECG PLOTS

```
figure('Name', 'ECG', 'Color', 'w', ...
    'Position', [50 50 1100 700]);

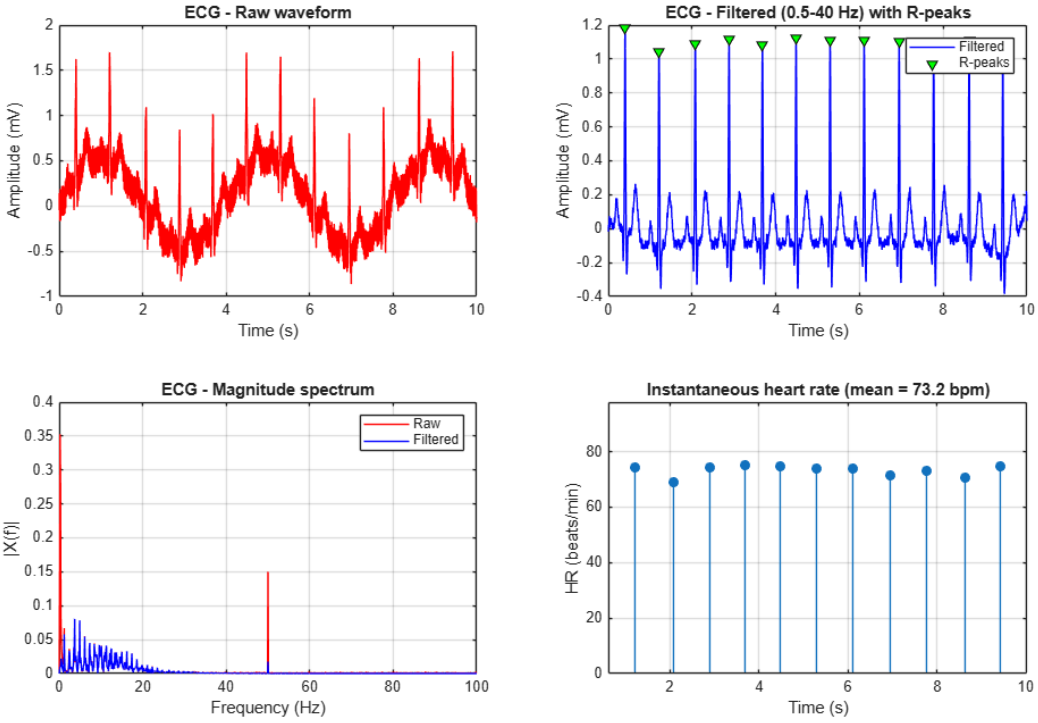
subplot(2,2,1);
plot(t_ecg, ecg_raw, 'r');
grid on;
title('ECG - Raw waveform');
xlabel('Time (s)');
ylabel('Amplitude (mV)');

subplot(2,2,2);
plot(t_ecg, ecg_f, 'b');
hold on;
plot(rPeakTimes, pk, 'kv', 'MarkerFaceColor', 'g');
grid on;
title(sprintf('ECG - Filtered (%.1f-%g Hz) with R-peaks', ...
    band_ecg));
xlabel('Time (s)');
ylabel('Amplitude (mV)');
legend('Filtered', 'R-peaks');

subplot(2,2,3);
plot(fE1, XE1, 'r');
hold on;
plot(fE1, XE2, 'b');
grid on;
xlim([0 100]);
title('ECG - Magnitude spectrum');
xlabel('Frequency (Hz)');
ylabel('|X(f)|');
legend('Raw', 'Filtered');

subplot(2,2,4);
stem(rPeakTimes(2:end), HR_all, 'filled');
grid on;
ylim([0 max(HR_all)*1.3]);
```

```
title(sprintf('Instantaneous heart rate (mean = %.1f bpm)', ...
    HR_avg));
xlabel('Time (s)');
ylabel('HR (beats/min)');
```



6. EEG PLOTS

```
figure('Name', 'EEG', 'Color', 'w', ...
    'Position', [80 80 1100 700]);

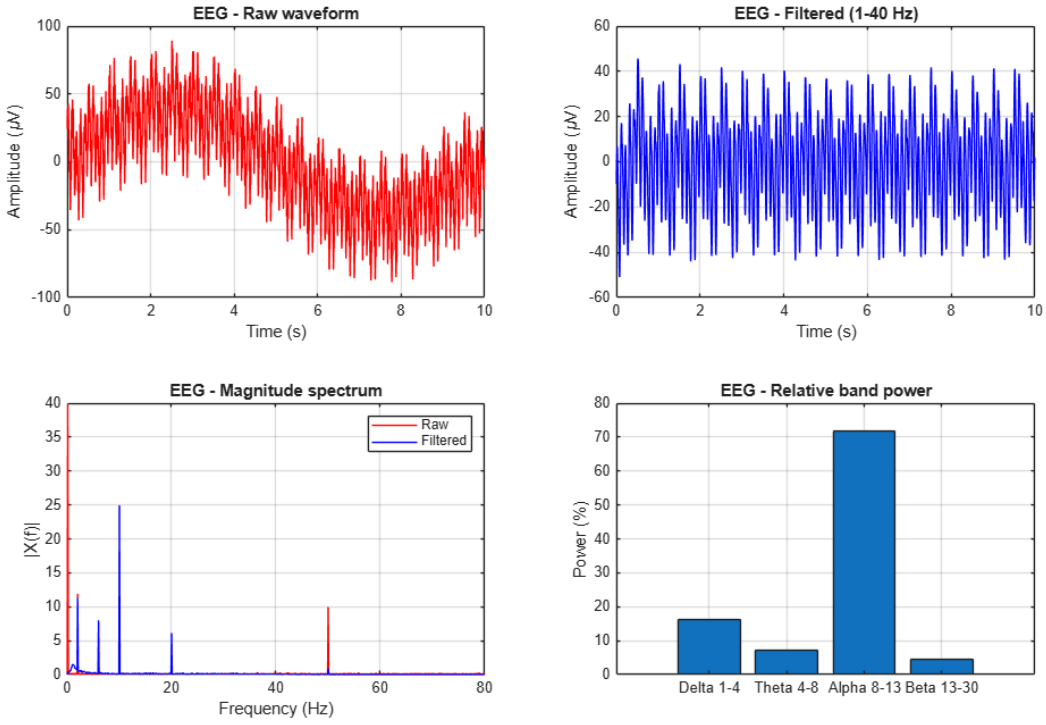
subplot(2,2,1);
plot(t_eeg, eeg_raw, 'r');
grid on;
title('EEG - Raw waveform');
xlabel('Time (s)');
ylabel('Amplitude (\muV)');

subplot(2,2,2);
plot(t_eeg, eeg_f, 'b');
grid on;
title(sprintf('EEG - Filtered (%g-%g Hz)', band_eeg));
xlabel('Time (s)');
ylabel('Amplitude (\muV)');

subplot(2,2,3);
plot(fG1, XG1, 'r');
hold on;
```

```
plot(fG1, XG2, 'b');
grid on;
xlim([0 80]);
title('EEG - Magnitude spectrum');
xlabel('Frequency (Hz)');
ylabel('|X(f)|');
legend('Raw', 'Filtered');

subplot(2,2,4);
bar(relPower);
set(gca, 'XTickLabel', ...
    {'Delta 1-4','Theta 4-8','Alpha 8-13','Beta 13-30'});
grid on;
title('EEG - Relative band power');
ylabel('Power (%)');
```



7. EMG PLOTS

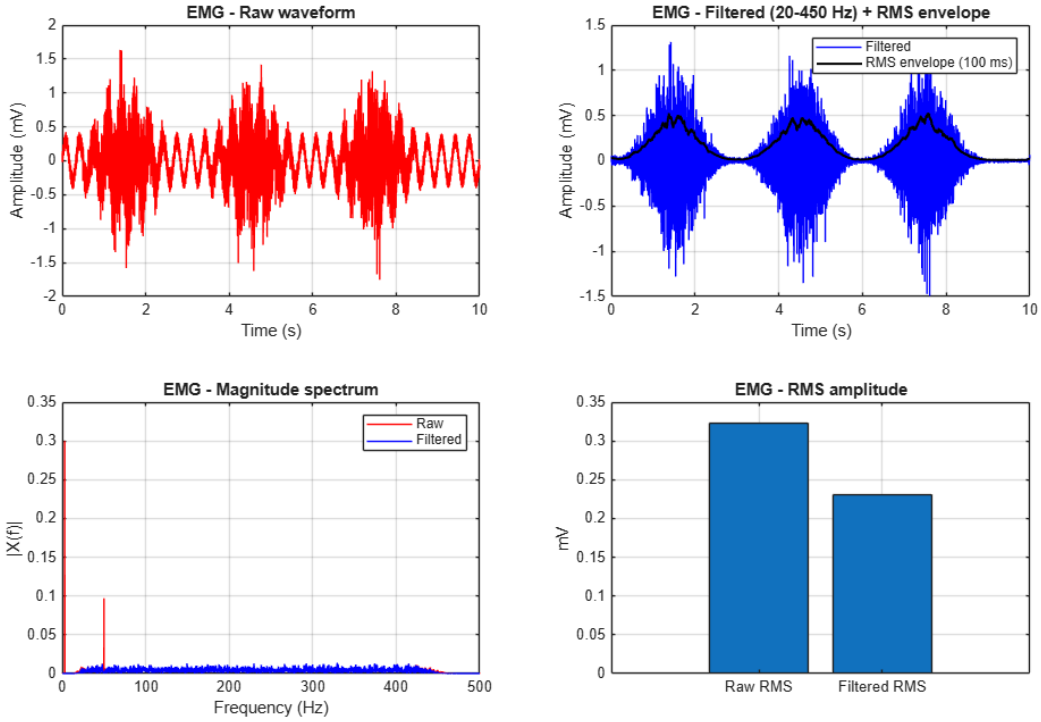
```
figure('Name', 'EMG', 'Color', 'w', ...
    'Position', [110 110 1100 700]);

subplot(2,2,1);
plot(t_emg, emg_raw, 'r');
grid on;
title('EMG - Raw waveform');
xlabel('Time (s)');
ylabel('Amplitude (mV)');
```

```
subplot(2,2,2);
plot(t_emg, emg_f, 'b');
hold on;
plot(t_emg, emg_env, 'k', 'LineWidth', 1.5);
grid on;
title(sprintf('EMG - Filtered (%g-%g Hz) + RMS envelope', ...
    band_emg));
xlabel('Time (s)');
ylabel('Amplitude (mV)');
legend('Filtered', 'RMS envelope (100 ms)');

subplot(2,2,3);
plot(fM1, XM1, 'r');
hold on;
plot(fM1, XM2, 'b');
grid on;
xlim([0 500]);
title('EMG - Magnitude spectrum');
xlabel('Frequency (Hz)');
ylabel('|X(f)|');
legend('Raw', 'Filtered');

subplot(2,2,4);
bar([RMS_raw RMS_f]);
set(gca, 'XTickLabel', {'Raw RMS','Filtered RMS'});
grid on;
title('EMG - RMS amplitude');
ylabel('mV');
```



8. FILTER FREQUENCY RESPONSES

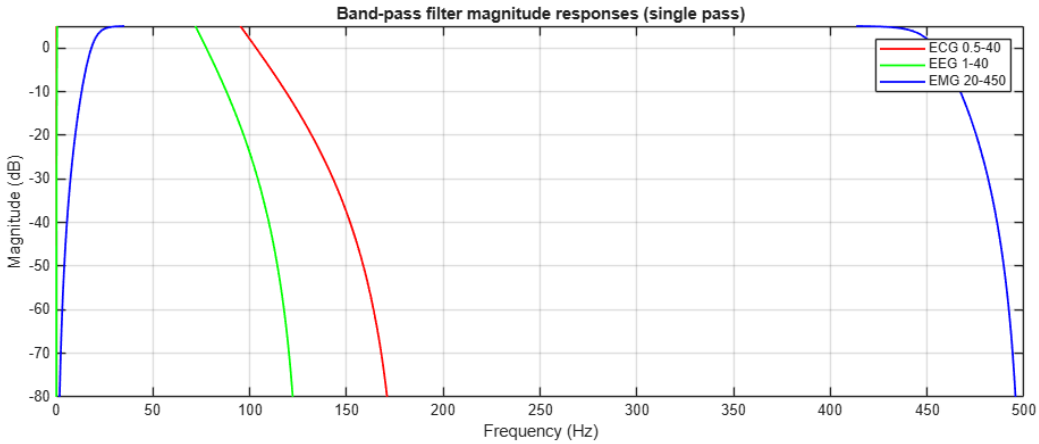
```
figure('Name', 'Filter responses', 'Color', 'w', ...
    'Position', [140 140 1100 400]);

[sE, gE] = bp_sos(band_ecg, fs_ecg, filtOrder);
[sG, gG] = bp_sos(band_eeg, fs_eeg, filtOrder);
[sM, gM] = bp_sos(band_emg, fs_emg, filtOrder);

[hE, wE] = freqz(sE, 4096, fs_ecg);
[hG, wG] = freqz(sG, 4096, fs_eeg);
[hM, wM] = freqz(sM, 4096, fs_emg);

plot(wE, 20*log10(abs(hE)+eps), 'r', ...
     wG, 20*log10(abs(hG)+eps), 'g', ...
     wM, 20*log10(abs(hM)+eps), 'b', ...
     'LineWidth', 1.3);

grid on;
xlim([0 500]);
ylim([-80 5]);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Band-pass filter magnitude responses (single pass)');
legend('ECG 0.5-40', 'EEG 1-40', 'EMG 20-450');
```



9. OUTPUT PARAMETER TABLE

```
fprintf('\n========== OUTPUT PARAMETER TABLE ==========\n');

fprintf('%-28s %-12s %-12s %-12s\n', ...
    'Parameter', 'ECG', 'EEG', 'EMG');

fprintf('%-28s %-12d %-12d %-12d\n', ...
    'Sampling freq (Hz)', fs_ecg, fs_eeg, fs_emg);

fprintf('%-28s %-12s %-12s %-12s\n', ...
    'Filter band (Hz)', ...
    sprintf('%g-%g', band_ecg), ...
    sprintf('%g-%g', band_eeg), ...
    sprintf('%g-%g', band_emg));

fprintf('%-28s %-12d %-12d %-12d\n', ...
    'Samples (10 s)', ...
    numel(ecg_raw), numel(eeg_raw), numel(emg_raw));

fprintf('%-28s %-12.2f %-12.2f %-12.2f\n', ...
    'SNR after filtering (dB)', SNR_ecg, SNR_eeg, SNR_emg);

fprintf('\n--- ECG features ---\n');
fprintf('Number of R-peaks : %d\n', numel(loc));
fprintf('Mean RR interval : %.3f s\n', mean(RR));
fprintf('Mean heart rate : %.1f beats/min\n', HR_avg);
fprintf('HR range (min / max) : %.1f / %.1f beats/min\n', ...
    min(HR_all), max(HR_all));

fprintf('\n--- EEG band power ---\n');

for k = 1:4
    fprintf('%-6s (%2d-%2d Hz) : %.3f uV^2 (%.1f %%)\n', ...
        bands(k).name, bands(k).range(1), ...
        bands(k).range(2), bandPower(k), relPower(k));
end
```

```
fprintf('Dominant rhythm : %s\n', bands(domIdx).name);

fprintf('\n--- EMG features ---\n');
fprintf('RMS (raw) : %.4f mV\n', RMS_raw);
fprintf('RMS (filtered) : %.4f mV\n', RMS_f);
fprintf('Peak RMS envelope : %.4f mV\n', max(emg_env));

if ~useRealData
    fprintf('\n--- Check against known clean synthetic signals ---\n');

    fprintf('ECG SNR raw : %.2f dB | filtered : %.2f dB\n', ...
        snr_db(ecg_clean, ecg_raw-ecg_clean), ...
        snr_db(ecg_clean, ecg_f-ecg_clean));

    fprintf('EEG SNR raw : %.2f dB | filtered : %.2f dB\n', ...
        snr_db(eeg_clean, eeg_raw-eeg_clean), ...
        snr_db(eeg_clean, eeg_f-eeg_clean));

    fprintf('EMG SNR raw : %.2f dB | filtered : %.2f dB\n', ...
        snr_db(emg_clean, emg_raw-emg_clean), ...
        snr_db(emg_clean, emg_f-emg_clean));
end
```

```
========== OUTPUT PARAMETER TABLE ==========
Parameter                    ECG          EEG          EMG
Sampling freq (Hz)           360          256          1000
Filter band (Hz)             0.5-40       1-40         20-450
Samples (10 s)               3600         2560         10000
SNR after filtering (dB)     -5.69        -2.90        0.27

--- ECG features ---
Number of R-peaks : 12
Mean RR interval : 0.820 s
Mean heart rate : 73.2 beats/min
HR range (min / max) : 69.0 / 75.3 beats/min

--- EEG band power ---
Delta  ( 1- 4 Hz) : 71.058 uV^2 (16.4 %)
Theta  ( 4- 8 Hz) : 31.701 uV^2 (7.3 %)
Alpha  ( 8-13 Hz) : 310.836 uV^2 (71.7 %)
Beta   (13-30 Hz) : 20.051 uV^2 (4.6 %)
Dominant rhythm : Alpha

--- EMG features ---
RMS (raw) : 0.3228 mV
RMS (filtered) : 0.2307 mV
Peak RMS envelope : 0.5220 mV

--- Check against known clean synthetic signals ---
ECG SNR raw : -4.55 dB | filtered : 7.59 dB
EEG SNR raw : -2.84 dB | filtered : 14.04 dB
EMG SNR raw : 0.35 dB | filtered : 20.98 dB
```

LOCAL FUNCTIONS

```
function [sos, g] = bp_sos(band, fs, order)
    [z, p, k] = butter(order, band/(fs/2), 'bandpass');
    [sos, g] = zp2sos(z, p, k);
end

function y = bp_filter(x, band, fs, order)
    [sos, g] = bp_sos(band, fs, order);
    y = filtfilt(sos, g, x);
end

function [f, X] = fft_spectrum(x, fs)
    N = numel(x);
    Xf = fft(x-mean(x));

    X = abs(Xf(1:floor(N/2)+1))/N;
    X(2:end-1) = 2*X(2:end-1);

    f = (0:floor(N/2))*fs/N;
end

function s = snr_db(sig, noise)
    s = 10*log10(sum(sig.^2)/sum(noise.^2));
end

% Synthetic ECG signal generator
function [x, clean] = synth_ecg(fs, T, mains)
    t = 0:1/fs:T-1/fs;
    clean = zeros(size(t));

    beatT = 0.4;

    while beatT < T
        % P, Q, R, S, T waves: amplitude, centre, width
        w = [ 0.15 -0.20 0.025;
             -0.15 -0.040 0.010;
              1.20  0.000 0.012;
             -0.25  0.040 0.012;
              0.30  0.250 0.050];

        for i = 1:5
            clean = clean + w(i,1)*exp( ...
                -((t-beatT-w(i,2)).^2)/(2*w(i,3)^2));
        end

        beatT = beatT + 0.83 + 0.03*randn;
    end

    base = 0.5*sin(2*pi*0.25*t);
    hum = 0.15*sin(2*pi*mains*t);
    noise = 0.03*randn(size(t));
```

```
    x = clean + base + hum + noise;
end

% Synthetic EEG signal generator
function [x, clean] = synth_eeg(fs, T, mains)
    t = 0:1/fs:T-1/fs;

    clean = 12*sin(2*pi*2*t+1) + ...
             8*sin(2*pi*6*t+2) + ...
            25*sin(2*pi*10*t) + ...
             6*sin(2*pi*20*t+0.5);

    clean = clean + 3*randn(size(t));

    drift = 40*sin(2*pi*0.1*t);
    hum = 10*sin(2*pi*mains*t);

    x = clean + drift + hum;
end

% Synthetic EMG signal generator
function [x, clean] = synth_emg(fs, T, mains)
    t = 0:1/fs:T-1/fs;
    n = randn(size(t));

    [sos, g] = bp_sos([20 450], fs, 4);
    n = filtfilt(sos, g, n);

    env = zeros(size(t));

    % Three muscle contractions
    for c = [1.5 4.5 7.5]
        env = env + exp(-((t-c).^2)/(2*0.5^2));
    end

    clean = 0.5*env.*n;

    motion = 0.3*sin(2*pi*3*t);
    hum = 0.1*sin(2*pi*mains*t);

    x = clean + motion + hum + 0.01*randn(size(t));
end
```

Published with MATLAB® R2026b